



9622 - A Complete Mapping of Stellar Evolution and the Interstellar Medium across M33

Cycle: 5, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Erik Rosolowsky (PI) (CSA Member)	University of Alberta
Dr. Adam Leroy (CoI) (CoPI) (US Admin CoI)	The Ohio State University
Dr. Sumit K Sarbadhicary (CoI) (CoPI)	The Johns Hopkins University
Dr. Julianne Dalcanton (CoI)	Flatiron Institute
Ms. Jaiden Peltonen (CoI) (CSA Member)	University of Alberta
Dr. Adam Smercina (CoI)	Space Telescope Science Institute
Dr. Eric Koch (CoI)	Smithsonian Institution Astrophysical Observatory
Dr. Benjamin F. Williams (CoI)	University of Washington
Dr. Thomas Williams (CoI) (ESA Member)	University of Manchester
Dr. Karin Marie Sandstrom (CoI)	University of California - San Diego
Dr. Ryan Chown (CoI)	Algoma University
Dr. Lent Clifton Johnson (CoI)	Northwestern University
Dr. Elizabeth Tarantino (CoI)	Space Telescope Science Institute
Mr. Tobin Wainer (CoI)	University of Washington
Dr. Francesco Belfiore (CoI) (ESA Member)	European Southern Observatory - Germany
Mr. Hamid Hassani (CoI) (CSA Member)	University of Alberta
Dr. David Thilker (CoI)	The Johns Hopkins University
Dr. Janice Lee (CoI)	Space Telescope Science Institute
Hannah Koziol (CoI)	University of California - San Diego
Dr. Zhuo Chen (CoI)	University of Washington
Devisree Tallapaneni (CoI)	The Ohio State University

Name	Institution
Dr. Adam G Ginsburg (CoI)	University of Florida
Mr. Ivan Sergeevich Gerasimov (CoI) (ESA Member)	Universite Cote d'Azur
Dr. Martha L. Boyer (CoI)	Space Telescope Science Institute
Dr. Mark R. Krumholz (CoI)	Australian National University
Dr. Thavisha Dharmawardena (CoI)	New York University
Harrisen Corbould (CoI) (CSA Member)	University of Alberta

OBSERVATIONS

Folder	Observation	Label	Observing Template	Science Target
M33				
	1	NIRCam Primary Mid	NIRCam Imaging	(1) M-33-Mortar1
	2	MIRI Off Observation	MIRI Imaging	(6) M33-OFF
	3	NIRCam Primary North	NIRCam Imaging	(2) M-33-Mortar2
	4	MIRI Off Observation	MIRI Imaging	(6) M33-OFF
	5	NIRCam Primary South	NIRCam Imaging	(3) M-33-Mortar3
	6	MIRI Off Observation	MIRI Imaging	(6) M33-OFF
	7	MIRI Primary Mid	MIRI Imaging	(5) M-33-Mortar5
	8	MIRI Off Observation	MIRI Imaging	(6) M33-OFF
	9	MIRI Primary North	MIRI Imaging	(4) M-33-Mortar4
	10	MIRI Off Observation	MIRI Imaging	(6) M33-OFF
	11	MIRI Primary South	MIRI Imaging	(7) M-33-South
	12	MIRI Off Observation	MIRI Imaging	(6) M33-OFF

ABSTRACT

We propose to map the central 33 kpc² of the nearby, star forming, disk galaxy M33 using NIRCam (6 bands) and MIRI (3 bands). This will be a first heavily resolved IR survey across a spiral galaxy. Combined with existing large investments by HST, ALMA, and more, this program will deliver a legacy data set for benchmarking the key physics of galactic evolution from the structure of the star forming interstellar medium (ISM), to the youngest stars, through dust production and the last stages of stellar evolution. We will use these observations to:

- (1) find >2200 massive young stellar objects (YSOs) over the star forming central disk of the galaxy, as a novel test for theories of star formation;

(2) use resolved, red stellar populations to probe the initial mass function and ancient star formation history of a spiral galaxy using stellar population synthesis;

(3) measure the contributions of O- and C-rich AGB stars to the dust budget of M33; and

(4) create high-quality maps of Paschen alpha plus massive YSOs and stars to provide a complete inventory of the sources of stellar feedback in the galaxy. We will then use polycyclic aromatic hydrocarbon (PAHs) to map out the small-scale structure of the ISM to find how feedback influences the gas around it.

We will deliver merged photometry of 40 million stars across 9 JWST + 6 HST bands and a curated mulitwavelength atlas of the nearest low-inclination disk galaxy to the Milky Way. With this JWST survey, M33 will form an essential case study to understand how galaxy evolution emerges from the collective action of millions of stars, HII regions, and gas clouds.

OBSERVING DESCRIPTION

Our primary observations consist of three NIRCam mosaics operated with MIRI in coordinated parallel mode. We use both modules of NIRCam with three filter configurations: F200W / F300W, F187N / F3335M and F115W / F444W. We use a FULLBOX dither pattern with 2 steps and 2 subpixel dithers optimized for MIRI parallels (observing F770W, F1000W, F2100w). Each mosaic is a 6 row x 1 column mosaic that we tile together. We introduce mosaic row and column offsets to tile the HST/PHATTER area at orientations when the program can be scheduled. This observing pattern accumulates a minimum of 300s in each NIRCam band and 277s in each MIRI parallel field. We have simulated the pointing resulting from these dither patterns for the primary and parallel instruments and find that most (>70%) of the field is covered with at least 2 times this integration depth.

Since the MIRI parallels in our primary observations cover disjoint parts of the galaxy, we two MIRI observations to fill in the gaps. Here, we use 4 point CYCLING patterns with FASTR1 readouts to accumulate 277s in F770W and F1000W and 566s in F2100W. One of the MIRI mosaics overlaps with archival data, so we have removed fields and bands of the mosaic that are already observed in the archival work. We operate NIRCam in parallel with these observations using 6 new bands (F150W/F405N, F090W/F360M, F212N/F430M).

We require a specific range of roll angles (80° to 110° in aperture PA) for the telescope to ensure that the MIRI parallels are simultaneously covering

JWST Proposal 9622 (Created: Tuesday, June 2, 2026, 8:00:21PM Eastern Standard Time) - Overview

the star forming part of the disk. This 30 degree range is selected to allow for a broad range of scheduling windows. Once scheduled, we will adjust row and column offsets for the specific PA of the telescope so that we can obtain ideal coverage at the given PA.

Since our MIRI data will be filled with diffuse emission, we include observations of a MIRI "OFF" field located outside of the galaxy to set the background levels. This is configured as a sequence with required observations within 1-2 days depending on the integration time in the associated MIRI observation.

Proposal 9622 - Targets - A Complete Mapping of Stellar Evolution and the Interstellar Medium across M33

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	M-33-Mortar1	RA: 01 33 58.2287 (23.4926196d) Dec: +30 41 30.32 (30.69176d) Equinox: J2000	Proper Motion RA: 5.479301949670298E-5 sec of time/yr Proper Motion Dec: 8.269999999999999E-4 arcsec/yr Epoch of Position: 2015.5	
	Comments: Category=Galaxy Description=[Disk galaxies, Spiral galaxies]				
	(2)	M-33-Mortar2	RA: 01 34 9.3667 (23.5390279d) Dec: +30 46 53.74 (30.78159d) Equinox: J2000	Proper Motion RA: 5.479301949670298E-5 sec of time/yr Proper Motion Dec: 8.269999999999999E-4 arcsec/yr Epoch of Position: 2015.5	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[Disk galaxies, Spiral galaxies]				
	(3)	M-33-Mortar3	RA: 01 33 49.5061 (23.4562754d) Dec: +30 35 10.22 (30.58617d) Equinox: J2000	Proper Motion RA: 5.479301949670298E-5 sec of time/yr Proper Motion Dec: 8.269999999999999E-4 arcsec/yr Epoch of Position: 2015.5	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[Disk galaxies, Spiral galaxies]				
	(4)	M-33-Mortar4	RA: 01 34 7.6141 (23.5317254d) Dec: +30 46 32.53 (30.77570d) Equinox: J2000	Proper Motion RA: 5.479301949670298E-5 sec of time/yr Proper Motion Dec: 8.269999999999999E-4 arcsec/yr Epoch of Position: 2015.5	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[Disk galaxies, Spiral galaxies]				
	(5)	M-33-Mortar5	RA: 01 33 55.3313 (23.4805471d) Dec: +30 39 5.24 (30.65146d) Equinox: J2000	Proper Motion RA: 5.479301949670298E-5 sec of time/yr Proper Motion Dec: 8.269999999999999E-4 arcsec/yr Epoch of Position: 2015.5	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[Disk galaxies, Spiral galaxies]				
	(6)	M33-OFF	RA: 01 36 31.9109 (24.1329621d) Dec: +30 33 47.61 (30.56322d) Equinox: J2000		
	Comments: Category=Calibration Description=[Telescope/sky background]				
	(7)	M-33-South	RA: 01 33 50.6592 (23.4610800d) Dec: +30 33 43.66 (30.56213d) Equinox: J2000	Proper Motion RA: 0.707 mas/yr Proper Motion Dec: 0.827 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Galaxy Description=[Spiral galaxies] Extended=YES				

Proposal 9622 - Observation 1 - A Complete Mapping of Stellar Evolution and the Interstellar Medium across M33

Observation	Proposal 9622, Observation 1: NIRCam Primary Mid										Wed Jun 03 01:00:21 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRCam Imaging											
	Coordinated Parallel Template(s): MIRI Imaging											
Diagnostics	(NIRCam Primary Mid (Obs 1)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.											
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 1:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 1:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 1:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 1:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
(NIRCam Primary Mid (Obs 1)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	M-33-Mortar1	RA: 01 33 58.2287 (23.4926196d) Dec: +30 41 30.32 (30.69176d) Equinox: J2000				Proper Motion RA: 5.479301949670298E-5 sec of time/yr Proper Motion Dec: 8.269999999999999E-4 arcsec/yr Epoch of Position: 2015.5					
	Comments: Category=Galaxy Description=[Disk galaxies, Spiral galaxies]											
Template	NIRCam Imaging					MIRI Imaging						
	Module: ALL					Subarray: FULL						
	Subarray: FULL											
	Target Placement: Module gap (large extended source)											
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	6	1	18.0		40.0		0.0		-30.0		DEFAULT	
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector		Dither Direct Images Primes
	1	FULLBOX		4				1		NIRCam Only		NO_DITHERING
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F187N	F335M	SHALLOW4	4	1		4	4	815.995	168019	
	2	F200W	F300M	SHALLOW4	3	1		4	4	601.259	168019	
	3	F115W	F444W	SHALLOW4	3	1		4	4	601.259	168019	

Proposal 9622 - Observation 1 - A Complete Mapping of Stellar Evolution and the Interstellar Medium across M33

Spectral Elements	MIRI Imaging	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F2100W	FASTR1	25	3	1		4	12	854.712	225860
	2	F1000W	FASTR1	25	2	1		4	8	566.108	225860
	3	F770W	FASTR1	25	2	1		4	8	566.108	225860
Special Requirements	Sequence Visits within 53.0 Days Aperture PA Range 80 to 110 Degrees (V3 80.07457694 to 110.07457694) Visits Same PA No Parallel Attachments Group Observations 1, 2 within 1 Days Same Aperture PA 1, 3, 5, 7, 9, 11 (V3 PAs differ)										

Proposal 9622 - Observation 2 - A Complete Mapping of Stellar Evolution and the Interstellar Medium across M33

Observation	Proposal 9622, Observation 2: MIRI Off Observation Wed Jun 03 01:00:21 GMT 2026									
	Diagnostic Status: Warning Observing Template: MIRI Imaging									
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous		
	(6)	M33-OFF	RA: 01 36 31.9109 (24.1329621d) Dec: +30 33 47.61 (30.56322d) Equinox: J2000							
Template	<i>Comments:</i> <i>Category=Calibration</i> <i>Description=[Telescope/sky background]</i>									
	Subarray FULL									
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
	1	CYCLING	1	3						DEFAULT
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time
	1	F2100W	FASTR1	25	3	1	Dither 1	3	9	641.034
	2	F1000W	FASTR1	25	2	1	Dither 1	3	6	424.581
	3	F770W	FASTR1	25	2	1	Dither 1	3	6	424.581
Special Requirements	Group Observations 1, 2 within 1 Days									

Proposal 9622 - Observation 3 - A Complete Mapping of Stellar Evolution and the Interstellar Medium across M33

Observation	Proposal 9622, Observation 3: NIRCam Primary North										Wed Jun 03 01:00:21 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRCam Imaging											
	Coordinated Parallel Template(s): MIRI Imaging											
Diagnostics	(NIRCam Primary North (Obs 3)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.											
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 3:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 3:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 3:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 3:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 3:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	(NIRCam Primary North (Obs 3)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(2)	M-33-Mortar2	RA: 01 34 9.3667 (23.5390279d) Dec: +30 46 53.74 (30.78159d) Equinox: J2000			Proper Motion RA: 5.479301949670298E-5 sec of time/yr Proper Motion Dec: 8.269999999999999E-4 arcsec/yr Epoch of Position: 2015.5						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	Category=Galaxy Description=[Disk galaxies, Spiral galaxies]											
Template	NIRCam Imaging											MIRI Imaging
	Module: ALL											Subarray: FULL
	Subarray: FULL											
	Target Placement: Module gap (large extended source)											
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	6	1	10.0		10.0		0.0		-30.0		DEFAULT	
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector		Dither Direct Images Primes
	1	FULLBOX		4				1		NIRCam Only		NO_DITHERING
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID		
	1	F187N	F335M	SHALLOW4	4	1	4	4	815.995	168019		
	2	F200W	F300M	SHALLOW4	3	1	4	4	601.259	168019		
	3	F115W	F444W	SHALLOW4	3	1	4	4	601.259	168019		

Proposal 9622 - Observation 3 - A Complete Mapping of Stellar Evolution and the Interstellar Medium across M33

Spectral Elements	MIRI Imaging	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F2100W	FASTR1	25	3	1		4	12	854.712	
	2	F1000W	FASTR1	25	2	1		4	8	566.108	
	3	F770W	FASTR1	25	2	1		4	8	566.108	
Special Requirements	Sequence Visits within 53.0 Days Aperture PA Range 80 to 110 Degrees (V3 80.07457694 to 110.07457694) Visits Same PA No Parallel Attachments Group Observations 3, 4 within 1 Days Same Aperture PA 1, 3, 5, 7, 9, 11 (V3 PAs differ)										

Proposal 9622 - Observation 4 - A Complete Mapping of Stellar Evolution and the Interstellar Medium across M33

Observation	Proposal 9622, Observation 4: MIRI Off Observation										Wed Jun 03 01:00:21 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(6)	M33-OFF	RA: 01 36 31.9109 (24.1329621d) Dec: +30 33 47.61 (30.56322d) Equinox: J2000								
	Comments: Category=Calibration Description=[Telescope/sky background]										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	3						DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F2100W	FASTR1	25	3	1	Dither 1	3	9	641.034	225860
	2	F1000W	FASTR1	25	2	1	Dither 1	3	6	424.581	225860
	3	F770W	FASTR1	25	2	1	Dither 1	3	6	424.581	225860
Special Requirements	Group Observations 3, 4 within 1 Days										

Proposal 9622 - Observation 5 - A Complete Mapping of Stellar Evolution and the Interstellar Medium across M33

Observation	Proposal 9622, Observation 5: NIRCam Primary South						Wed Jun 03 01:00:21 GMT 2026
	Diagnostic Status: Warning						
	Observing Template: NIRCam Imaging						
	Coordinated Parallel Template(s): MIRI Imaging						
Diagnostics	(NIRCam Primary South (Obs 5)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.						
	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 5:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 5:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 5:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 5:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 5:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 5:7) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 5:8) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 5:9) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 5:10) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 5:11) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 5:12) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 5:13) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 5:14) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
Fixed Targets	(NIRCam Primary South (Obs 5)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.						
	#	Name	Target Coordinates	Targ. Coord. Corrections		Miscellaneous	
	(3)	M-33-Mortar3	RA: 01 33 49.5061 (23.4562754d) Dec: +30 35 10.22 (30.58617d) Equinox: J2000	Proper Motion RA: 5.479301949670298E-5 sec of time/yr Proper Motion Dec: 8.269999999999999E-4 arcsec/yr Epoch of Position: 2015.5			
Template	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.						
	Category=Galaxy						
	Description=[Disk galaxies, Spiral galaxies]						
Mosaic	NIRCam Imaging		MIRI Imaging				
	Module: ALL		Subarray: FULL				
	Subarray: FULL						
Dithers	Target Placement: Module gap (large extended source)						
	Rows	Columns	Row Overlap %	Column Overlap %	Row shift (deg)	Column shift (deg)	Tile Order
Dithers	7	2	18.0	65.0	0.0	-30.0	DEFAULT
	#	Primary Dither Type	Primary Dithers	Dither Size	Subpixel Positions	Coordinated Parallel Subpixel Selector	Dither Direct Images Primes
Dithers	1	FULLBOX	4		1	NIRCam Only	NO_DITHERING

Proposal 9622 - Observation 5 - A Complete Mapping of Stellar Evolution and the Interstellar Medium across M33

Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F187N	F300M	SHALLOW4	4	1	4	4	815.995	168019	
	2	F200W	F335M	SHALLOW4	3	1	4	4	601.259	168019	
	3	F115W	F444W	SHALLOW4	3	1	4	4	601.259	168019	
Spectral Elements	MIRI Imaging	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F2100W	FASTR1	25	3	1		4	12	854.712	225860
	2	F1000W	FASTR1	25	2	1		4	8	566.108	225860
	3	F770W	FASTR1	25	2	1		4	8	566.108	225860
Special Requirements	Sequence Visits within 53.0 Days Aperture PA Range 80 to 110 Degrees (V3 80.07457694 to 110.07457694) Visits Same PA No Parallel Attachments										
	Group Observations 5, 6 within 1 Days Same Aperture PA 1, 3, 5, 7, 9, 11 (V3 PAs differ)										

Proposal 9622 - Observation 6 - A Complete Mapping of Stellar Evolution and the Interstellar Medium across M33

Observation	Proposal 9622, Observation 6: MIRI Off Observation										Wed Jun 03 01:00:21 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(6)	M33-OFF	RA: 01 36 31.9109 (24.1329621d) Dec: +30 33 47.61 (30.56322d) Equinox: J2000								
	Comments: Category=Calibration Description=[Telescope/sky background]										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	3						DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F2100W	FASTR1	25	3	1	Dither 1	3	9	641.034	225860
	2	F1000W	FASTR1	25	2	1	Dither 1	3	6	424.581	225860
	3	F770W	FASTR1	25	2	1	Dither 1	3	6	424.581	225860
Special Requirements	Group Observations 5, 6 within 1 Days										

Proposal 9622 - Observation 7 - A Complete Mapping of Stellar Evolution and the Interstellar Medium across M33

Observation	Proposal 9622, Observation 7: MIRI Primary Mid						Wed Jun 03 01:00:21 GMT 2026	
	Diagnostic Status: Warning							
	Observing Template: MIRI Imaging							
	Coordinated Parallel Template(s): NIRCam Imaging							
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
	(Visit 7:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
	(Visit 7:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
	(Visit 7:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
	(Visit 7:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
	(Visit 7:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
	(Visit 7:7) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
	(Visit 7:8) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
	(Visit 7:9) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
	(Visit 7:10) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
	(Visit 7:11) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
	(Visit 7:12) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
	(Visit 7:13) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
	(Visit 7:14) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
	(Visit 7:15) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
	(Visit 7:16) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
	(Visit 7:17) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
	(Visit 7:18) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
	(Visit 7:19) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
	(Visit 7:20) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
	(Visit 7:21) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
	(Visit 7:22) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
	(Visit 7:23) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
	(Visit 7:24) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
	(MIRI Primary Mid (Obs 7)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.							
	Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections		Miscellaneous	
		(5)	M-33-Mortar5	RA: 01 33 55.3313 (23.4805471d) Dec: +30 39 5.24 (30.65146d) Equinox: J2000	Proper Motion RA: 5.479301949670298E-5 sec of time/yr Proper Motion Dec: 8.269999999999999E-4 arcsec/yr Epoch of Position: 2015.5			
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.								
Category=Galaxy Description=[Disk qalaxies, Spiral qalaxies]								
Template	MIRI Imaging			NIRCam Imaging				
	Subarray: FULL			Module: ALL				
				Subarray: FULL				
Mosaic	Rows	Columns	Row Overlap %	Column Overlap %	Row shift (deg)	Column shift (deg)	Tile Order	
	6	4	10.0	10.0	0.0	0.0	HILBERT_CURVE	

Proposal 9622 - Observation 7 - A Complete Mapping of Stellar Evolution and the Interstellar Medium across M33

Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4						DEFAULT	
Spectral Elements	MIRI Imaging	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F2100W	FASTR1	25	2	1	Dither 1	4	8	566.108	225860
	2	F1000W	FASTR1	25	1	1	Dither 1	4	4	277.504	225860
	3	F770W	FASTR1	25	1	1	Dither 1	4	4	277.504	225860
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F212N	F405N+F444W	SHALLOW4	2	1	4	4	386.524		
	2	F150W	F430M	SHALLOW4	1	1	4	4	171.788		
	3	F090W	F360M	SHALLOW4	1	1	4	4	171.788		
Special Requirements	Group Visits within 53.0 Days Aperture PA Range 80 to 110 Degrees (V3 75.16455103 to 105.16455103) Visits Same PA No Parallel Attachments										
	Group Observations 7, 8 within 2 Days Same Aperture PA 1, 3, 5, 7, 9, 11 (V3 PAs differ)										

Proposal 9622 - Observation 8 - A Complete Mapping of Stellar Evolution and the Interstellar Medium across M33

Observation	Proposal 9622, Observation 8: MIRI Off Observation Wed Jun 03 01:00:21 GMT 2026									
	Diagnostic Status: Warning Observing Template: MIRI Imaging									
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous		
	(6)	M33-OFF	RA: 01 36 31.9109 (24.1329621d) Dec: +30 33 47.61 (30.56322d) Equinox: J2000							
Template	<i>Comments:</i> <i>Category=Calibration</i> <i>Description=[Telescope/sky background]</i>									
	Subarray FULL									
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
	1	CYCLING	1	4						DEFAULT
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time
	1	F2100W	FASTR1	25	2	1	Dither 1	4	8	566.108
	2	F1000W	FASTR1	25	1	1	Dither 1	4	4	277.504
	3	F770W	FASTR1	25	1	1	Dither 1	4	4	277.504
Special Requirements	Group Observations 7, 8 within 2 Days									

Proposal 9622 - Observation 9 - A Complete Mapping of Stellar Evolution and the Interstellar Medium across M33

Observation	Proposal 9622, Observation 9: MIRI Primary North Diagnostic Status: Warning Observing Template: MIRI Imaging Coordinated Parallel Template(s): NIRCam Imaging Wed Jun 03 01:00:21 GMT 2026													
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:2) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:3) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:4) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:5) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:6) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:7) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:8) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:9) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:10) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:11) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:12) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:13) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:14) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:15) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:16) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:17) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:18) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:19) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:20) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:21) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:22) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:23) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:24) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:25) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:26) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:27) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:28) Warning (Form): Overheads are provisional until the Visit Planner has been run. (MIRI Primary North (Obs 9)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.													
Fixed Targets	<table> <tr> <th>#</th><th>Name</th><th>Target Coordinates</th><th>Targ. Coord. Corrections</th><th>Miscellaneous</th></tr> <tr> <td>(4)</td><td>M-33-Mortar4</td><td> RA: 01 34 7.6141 (23.5317254d) Dec: +30 46 32.53 (30.77570d) Equinox: J2000 </td><td> Proper Motion RA: 5.479301949670298E-5 sec of time/yr Proper Motion Dec: 8.269999999999999E-4 arcsec/yr Epoch of Position: 2015.5 </td><td></td></tr> </table> Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[Disk galaxies, Spiral galaxies]				#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous	(4)	M-33-Mortar4	RA: 01 34 7.6141 (23.5317254d) Dec: +30 46 32.53 (30.77570d) Equinox: J2000	Proper Motion RA: 5.479301949670298E-5 sec of time/yr Proper Motion Dec: 8.269999999999999E-4 arcsec/yr Epoch of Position: 2015.5	
#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous										
(4)	M-33-Mortar4	RA: 01 34 7.6141 (23.5317254d) Dec: +30 46 32.53 (30.77570d) Equinox: J2000	Proper Motion RA: 5.479301949670298E-5 sec of time/yr Proper Motion Dec: 8.269999999999999E-4 arcsec/yr Epoch of Position: 2015.5											

Proposal 9622 - Observation 9 - A Complete Mapping of Stellar Evolution and the Interstellar Medium across M33

Template	MIRI Imaging					NIRCam Imaging					
	Subarray: FULL					Module: ALL					
						Subarray: FULL					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)	Column shift (deg)		Tile Order	
	7	4	10.0		10.0		0.0	0.0		HILBERT_CURVE	
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4						DEFAULT	
Spectral Elements	MIRI Imaging	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F2100W	FASTR1	25	2	1	Dither 1	4	8	566.108	225860
	2	F1000W	FASTR1	25	1	1	Dither 1	4	4	277.504	225860
	3	F770W	FASTR1	25	1	1	Dither 1	4	4	277.504	225860
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F212N	F405N+F444W	SHALLOW4	2	1	4	4	386.524		
	2	F150W	F430M	SHALLOW4	1	1	4	4	171.788		
	3	F090W	F360M	SHALLOW4	1	1	4	4	171.788		
Special Requirements	Group Visits within 53.0 Days Aperture PA Range 80 to 110 Degrees (V3 75.16455103 to 105.16455103) Visits Same PA No Parallel Attachments										
	Group Observations 9, 10 within 2 Days Same Aperture PA 1, 3, 5, 7, 9, 11 (V3 PAs differ)										

Proposal 9622 - Observation 10 - A Complete Mapping of Stellar Evolution and the Interstellar Medium across M33

Observation	Proposal 9622, Observation 10: MIRI Off Observation										Wed Jun 03 01:00:21 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(6)	M33-OFF	RA: 01 36 31.9109 (24.1329621d) Dec: +30 33 47.61 (30.56322d) Equinox: J2000								
	Comments: Category=Calibration Description=[Telescope/sky background]										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4						DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F2100W	FASTR1	25	2	1	Dither 1	4	8	566.108	225860
	2	F1000W	FASTR1	25	1	1	Dither 1	4	4	277.504	225860
	3	F770W	FASTR1	25	1	1	Dither 1	4	4	277.504	225860
Special Requirements	Group Observations 9, 10 within 2 Days										

Proposal 9622 - Observation 11 - A Complete Mapping of Stellar Evolution and the Interstellar Medium across M33

Observation	Proposal 9622, Observation 11: MIRI Primary South						Wed Jun 03 01:00:21 GMT 2026
	Diagnostic Status: Warning Observing Template: MIRI Imaging						
Diagnostics	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 11:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 11:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 11:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 11:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 11:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 11:7) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 11:8) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 11:9) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 11:10) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 11:11) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 11:12) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 11:13) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 11:14) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 11:15) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 11:16) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 11:17) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 11:18) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 11:19) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(Visit 11:20) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
	(MIRI Primary South (Obs 11)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.						
Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous		
	(7)	M-33-South	RA: 01 33 50.6592 (23.4610800d) Dec: +30 33 43.66 (30.56213d) Equinox: J2000	Proper Motion RA: 0.707 mas/yr Proper Motion Dec: 0.827 mas/yr Epoch of Position: 2000			
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.						
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.						
	Category=Galaxy						
	Description=[Spiral galaxies]						
	Extended=YES						
Template	Subarray						
	FULL						
Mosaic	Rows	Columns	Row Overlap %	Column Overlap %	Row shift (deg)	Column shift (deg)	Tile Order
	5	6	10.0	10.0	0.0	0.0	HILBERT_CURVE

Proposal 9622 - Observation 11 - A Complete Mapping of Stellar Evolution and the Interstellar Medium across M33

Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4						DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	25	1	1	Dither 1	4	4	277.504	
Special Requirements	Group Visits within 53.0 Days Aperture PA Range 80 to 110 Degrees (V3 75.16455103 to 105.16455103) Visits Same PA										
	Group Observations 11, 12 within 1 Days Same Aperture PA 1, 3, 5, 7, 9, 11 (V3 PAs differ)										

Proposal 9622 - Observation 12 - A Complete Mapping of Stellar Evolution and the Interstellar Medium across M33

Observation	Proposal 9622, Observation 12: MIRI Off Observation										Wed Jun 03 01:00:21 GMT 2026
	Diagnostic Status: Warning										
Diagnostics	Observing Template: MIRI Imaging										
	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(6)	M33-OFF	RA: 01 36 31.9109 (24.1329621d) Dec: +30 33 47.61 (30.56322d) Equinox: J2000								
Template	Comments: Category=Calibration Description=[Telescope/sky background]										
	Subarray FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4						DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	25	1	1	Dither 1	4	4	277.504	225860
Special Requirements	Group Observations 11, 12 within 1 Days										